

Problem-Solution Fit

Date	02 July 2026
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] • Credit card applicants • Banks and financial institutions • Credit analysts • Risk managers	6. CUSTOMER LIMITATIONS [CL] • Limited understanding of approval criteria • Incomplete financial information • Different credit histories • Limited access to banking experts	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Manual Review: Accurate but slow and time-consuming. Rule-Based Systems: Fast but less flexible. ML-Based Prediction: Faster, scalable, and more accurate but requires quality training data.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Manual approval process is slow (High) • Human errors in decision-making (High) • Difficulty identifying high-risk applicants (High) • Long waiting time for customers (Medium)	9. PROBLEM ROOT / CAUSE [RC] • Manual verification takes time • Multiple financial factors influence approval • Large volume of applications • Inconsistent evaluation methods • Human bias in decision-making	7. BEHAVIOR + ITS INTENSITY [BE] • Customers check eligibility before applying (High) • Banks verify applicant data (High) • Customers expect quick responses (Medium)

3. TRIGGERS TO ACT [TR]

- Need for faster approval decisions
- Increase in credit card applications
- **Reduce financial risk**
- **Improve customer satisfaction**

4. EMOTIONS BEFORE / AFTER [EM]

Before: Anxious, uncertain, frustrated.

After: Confident, satisfied, relieved due to quick and accurate predictions.

10. YOUR SOLUTION [SL]

Develop a **Machine Learning-based Credit Card Approval Prediction System** using **Logistic Regression, Decision Tree, Random Forest, and XGBoost**. The system preprocesses applicant data, predicts approval or rejection, and provides instant results through a **Flask web application** deployed on **IBM Watson Machine Learning**, improving accuracy, reducing processing time, and supporting reliable decision-making.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE:

Flask Web Application,
Bank Portal, IBM Cloud

OFFLINE:

Bank branches, Customer
service centers.